# ManologAI: Unified Personal Life Analytics and Behavioral Insight Engine

**Minor Project (ITUGPV1) Report**

Information Technology, 7th Sem

**Submitted by**

**Brijesh Kumar, Narendra, Suraj Biswas**

**(23036115, 23036134, 23036168)**

**Under the supervision of**

**Dr. Amit Kumar Khaskalam**




**To**

**Department of Information Technology**

**Guru Ghasidas Vishwavidyalaya**

**Session 2026-2027**

# DECLARATION

We certify that

(a) The work contained in this report has been done by us under the guidance of our supervisor.

(b) The work has not been submitted to any other Institute for any curriculum.

(c) We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

We hereby declare that the project entitled – "**ManologAI: Unified Personal Life Analytics and Behavioral Insight Engine**", which is being submitted as a Project of **Information Technology, 7th Sem** to Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.) is an authentic record of our genuine work done under the guidance of **Dr. Amit Kumar Khaskalam**, Associate Professor, Department of Information Technology, SOS Engineering and Technology, Guru Ghasidas Vishwavidyalaya (A Central University), Bilaspur, Chhattisgarh.

Date: 30, Oct 2027 (Brijesh Kumar, Narendra, Suraj Biswas)
Place: Bilaspur (23036115, 23036134, 23036168)

**DEPARTMENT OF INFORMATION TECHNOLOGY**

**GURU GHASIDAS VISHWAVIDYALAYA**

**BILASPUR - 495009, INDIA**




# CERTIFICATE

This is to certify that the project report entitled **ManologAI: Unified Personal Life Analytics and Behavioral Insight Engine** submitted by **Brijesh Kumar, Narendra, Suraj Biswas** (Roll No. 23036115, 23036134, 23036168) to Guru Ghasidas Vishwavidyalaya, is a record of bonafide work carried out by them as a part of the curriculum requirements of the Information Technology, 7th Sem Project under my supervision and guidance.

Amit Kumar Khaskalam

Date: 30, Oct 2027
Place: Bilaspur

# Project Approval Sheet

The project entitled **ManologAI: Unified Personal Life Analytics and Behavioral Insight Engine** submitted by **Brijesh Kumar, Narendra, Suraj Biswas** is submitted in partial fulfillment of the requirements for the **7th Semester Project** of the Bachelor of Technology in Information Technology at Guru Ghasidas Vishwavidyalaya, Bilaspur, and is hereby approved.

**Internal Examiner**

Dr. Amit Kumar Khaskalam

**External Examinee**

1. Dr. Amit Kumar Khaskalam
2. Mr. Suhel Ahamed
3. Mr. Bratin Chakrabarti

**Dr. Manoj Kumar**

Head of the Department of Information Technology

Guru Ghasidas Vishwavidyalaya

# Acknowledgements

We express our sincere gratitude to **Dr. Amit Kumar Khaskalam**, Associate Professor, Department of Information Technology, who served as the **Project Guide** for this project. We are deeply thankful for his valuable guidance, continuous encouragement, constructive suggestions, and constant supervision throughout the course of this work.

We would like to express our sincere thanks to the **Project In-Charge**, **Mr. Deepak Kant Netam**, Assistant Professor, Department of Information Technology, for his guidance, coordination, and support during the execution of this project.

We would also like to express our heartfelt gratitude to the Head of the Department of Information Technology, **Dr. Manoj Kumar**, for his leadership, motivation, and for providing the necessary academic support and resources during the development of this project.

We are extremely thankful to **Prof. M. Chakradhara Rao**, Dean, School of Studies in Engineering and Technology, Guru Ghasidas Vishwavidyalaya, for providing the required infrastructural facilities and a conducive academic environment, without which this work would not have been possible.

We also wish to extend our sincere thanks to the faculty members of the Department of Information Technology and our classmates for their inspiration, encouragement, insightful comments, and constructive suggestions, which helped in improving the quality of this project work.

# Abstract

---

Name of the student: **Brijesh Kumar, Narendra, Suraj Biswas**

Roll No: **23036115, 23036134, 23036168**

Curriculum for which submitted: **Information Technology, 7th Sem, Project**

Degree: **Bachelor Of Technology**

Department: **Department of Information Technology**

Project Name: **ManologAI: Unified Personal Life Analytics and Behavioral Insight Engine**

Project Supervisor: **Dr. Amit Kumar Khaskalam**

Month and year of thesis submission: **30, Oct 2027**

---

In modern digital productivity, personal life data is fundamentally fragmented across isolated silos such as disparate journaling tools, habit trackers, and health applications, preventing users from understanding cross-domain behavioral patterns. This project presents **ManologAI**, an engineering-grade, privacy-first personal life analytics platform and behavioral insight engine. Designed to resolve the fragmented "Quantified-Self" problem, ManologAI unites qualitative daily reflections and an 8-scale quantized emotion tracking system with quantitative lifestyle metrics (sleep duration, screen time, study hours, habits, expenses, and weekly timetables) into a single cohesive data pipeline.

Unlike conventional platforms relying on non-deterministic generative models that risk hallucination, ManologAI incorporates a deterministic behavioral intelligence engine utilizing Pearson correlation coefficient ($r$) analysis and counterfactual baseline evaluation ($\Delta$) to reveal empirical cause-and-effect relationships between daily

routines and psychological valence. To guarantee data integrity and eliminate retrospective falsification, the platform enforces a strict 24-hour temporal integrity lock. Built on a decoupled four-tier architecture featuring React 19, Vite, Tailwind CSS, Node.js, Express 5, and MongoDB with standards-compliant RFC 8291/8292 VAPID Web Push notifications, ManologAI delivers a frictionless sub-two-minute daily logging ritual, robust data integrity, and actionable personal intelligence for lifestyle optimization.

# Contents

**Declaration** i

**Certificate** ii

**Project Approval Sheet** iii

**Acknowledgements** iv

**Abstract** v

**Contents** vii

**List of Figures** ix

**1 Introduction** 1
- 1.1 Aim & Objectives . . . 3

**2 Literature Review** 5
- 2.1 Existing Systems and Tracking Paradigms . . . 5
- 2.2 Limitations of Existing Systems . . . 6
- 2.3 Advantages of the Proposed ManologAI Platform . . . 8

**3 System Analysis** 9
- 3.1 Requirement Analysis . . . 9
- 3.2 Functional and Non-Functional Requirements . . . 10
  - 3.2.1 Functional Requirements . . . 10
  - 3.2.2 Non-Functional Requirements . . . 12

**4 System Design** 13
- 4.1 System Architecture . . . 13
- 4.2 Data Flow Pipelines . . . 14
- 4.3 UML and Database Schema Modeling . . . 15

**5 Technology Stack and Tools** 18

5.1 Programming Languages and Runtime . . . . . 18
5.2 Frontend Technologies and Frameworks . . . . . 19
5.3 Backend Engine and Security Middleware . . . . . 19
5.4 Database and Cloud Infrastructure . . . . . 20
5.5 Developer Tools and Quality Verification . . . . . 21
5.6 Hardware and Software Requirements . . . . . 21

**6 Implementation & Mathematical Modeling 22**
6.1 Behavioral Intelligence & Mathematical Formulations . . . . . 22
6.1.1 Discrete Emotion Quantization Matrix . . . . . 23
6.1.2 Pearson Correlation Coefficient Engine . . . . . 23
6.1.3 Counterfactual Habit Impact Engine . . . . . 24
6.1.4 Dynamic Habit Consistency Scoring and Ingestion Constraints 25
6.2 Module-Wise Implementation . . . . . 26
6.2.1 Workspace & Central Dashboard Module . . . . . 26
6.2.2 Authentication & Multi-Device Session Module . . . . . 26
6.2.3 Habit Management & Scheduling Module . . . . . 28
6.2.4 Behavioral Analytics & Visualizations Module . . . . . 29
6.2.5 Life Ratings & Evaluation Dashboard Module . . . . . 30
6.3 Progressive Web App & VAPID Web Push Implementation . . . . . 31
6.4 Testing and Quality Verification . . . . . 32

**7 Conclusion and Future Work 33**
7.1 Conclusion . . . . . 33
7.2 Future Enhancements . . . . . 34

**Bibliography 35**

# List of Figures

4.1 Data Flow Diagram . . . . . 15
4.2 UML Diagram . . . . . 17

6.1 Workspace (Main Dashboard) . . . . . 26
6.2 Sign In . . . . . 27
6.3 Sign Up . . . . . 27
6.4 Profile Page . . . . . 28
6.5 Hacbit Tracking . . . . . 29
6.6 Analytics Visualizations . . . . . 30
6.7 Life Ratings . . . . . 31

# Chapter 1

# Introduction

In the contemporary digital landscape, personal productivity, mental health, and lifestyle optimization have emerged as paramount concerns. The widespread adoption of smart devices and digital trackers has spurred the growth of the "Quantified-Self" movement—an empirical paradigm where individuals continuously log behavioral, physiological, and psychological metrics to gain actionable self-knowledge. However, despite the abundance of specialized software applications, users face significant systemic fragmentation. Qualitative reflections (such as daily diary writing and mood tracking) are fundamentally decoupled from quantitative routines (such as sleep monitoring, screen time tracking, study sessions, and habit consistency).

Modern users frequently employ multiple disjointed applications to track their daily lives: dedicated note-taking tools for journaling, separate habit trackers for recurring rituals, specialized affect apps for mood logs, and standalone timers or spreadsheets for study and financial records. These isolated repositories of personal data create telemetry silos. Crucially, disjointed architectures cannot perform cross-domain correlation. They fail to address vital multidimensional lifestyle questions, such as the

direct empirical impact of inadequate sleep on daily emotional valence, the statistical influence of habitual exercise on career satisfaction, or the weekly cyclicality of psychological stress.

Furthermore, traditional self-tracking tools suffer from severe retrospective bias and cognitive friction. Most mobile trackers permit arbitrary historical backfilling, allowing individuals to retroactively log or falsify entries days later to maintain artificial streaks. This historical revisionism distorts longitudinal datasets and degrades the validity of any analytical inference. Concurrently, switching between multiple single-purpose applications imposes high cognitive load, causing a vast majority of users to abandon tracking within a few weeks.

To overcome these structural limitations, this project presents **ManologAI**—an engineering-grade, privacy-first personal life analytics platform and behavioral insight engine. Designed specifically to resolve the fragmented Quantified-Self problem, ManologAI unites qualitative daily reflections and an 8-scale quantized emotion tracking system with quantitative lifestyle metrics (sleep duration, screen time, study hours, habits, daily expenses, and weekly timetables) into a single cohesive, high-performance data pipeline.

Rather than relying on non-deterministic generative models that are susceptible to analytical hallucination and opaque inference, ManologAI incorporates a deterministic behavioral intelligence engine based on statistical regression. By computing Pearson correlation coefficients ($r$) across sliding temporal windows and conducting empirical counterfactual baseline analyses ($\Delta$), ManologAI transforms raw chronological logs into objective, explainable, and natural-language diagnostic feedback. To ensure pristine historical datasets, the platform enforces a strict 24-hour temporal

integrity lock via API middleware, precluding future logging and retroactive falsification. Built as a responsive Progressive Web App (PWA) with native RFC 8291/8292 VAPID push notifications, ManologAI delivers a frictionless sub-two-minute daily logging ritual, robust data integrity, and actionable personal intelligence for lifestyle optimization.

## 1.1 Aim & Objectives

The primary aim of this project is to design, implement, and evaluate **ManologAI**, a comprehensive personal life analytics platform and deterministic behavioral insight engine that unifies qualitative and quantitative behavioral telemetry within an ergonomic, secure, and privacy-first full-stack environment.

The specific objectives of this project are:

- To architect and develop a responsive, cross-platform Progressive Web App (PWA) using React 19, Vite, and Tailwind CSS that enables a comprehensive daily check-in ritual in under two minutes.
- To construct a high-throughput, decoupled RESTful backend application using Node.js and Express 5, implementing modular domain routers and robust middleware pipelines.
- To design a schema-enforced relational document store using MongoDB and Mongoose 9, incorporating compound unique indexing to eliminate race conditions and enforce transactional consistency.

- To formulate and implement a deterministic behavioral intelligence engine executing Pearson correlation coefficient ($r$) analysis and counterfactual baseline evaluation ($\Delta$) to reveal empirical cause-and-effect relationships.
- To enforce strict data integrity through a 24-hour temporal integrity lock mechanism that rejects future submissions and write-locks historical records.
- To deploy a standards-compliant Voluntary Application Server Identification (VAPID) Web Push notification daemon with an automated background cron worker, bypassing third-party tracking SDKs.
- To build interactive data visualization interfaces utilizing Recharts for multi-axis radar charts, scatter plots, and longitudinal time-series analysis.
- To rigorously verify application robustness, security, and algorithmic reliability through an automated 16-scenario end-to-end integration test suite.

# Chapter 2

# Literature Review

## 2.1 Existing Systems and Tracking Paradigms

With the accelerated digitization of daily life, numerous personal tracking systems have emerged across commercial and research domains. Existing lifestyle tracking software can broadly be classified into four primary categories:

**1. Digital Journaling and Note-Taking Platforms:** Applications such as Notion, Day One, and Journey provide expressive rich-text editors for subjective daily reflections, gratitude journaling, and multimedia logging. While effective for qualitative capture, these platforms lack structured numerical schemas, automated metric validation, and mathematical telemetry processing. Analyzing relationships between recorded sentiments and objective variables requires tedious manual exporting and external computational tools.

**2. Dedicated Habit Tracking Applications:** Tools such as Streaks, Habitica, and Loop Habit Tracker focus exclusively on recurring binary or quantitative

habits (e.g., daily reading, gym sessions, hydration). They incorporate gamification mechanics, streak counters, and completion percentages. However, they operate in complete isolation from the user's broader daily context, offering no insight into how habit completion affects overall subjective well-being, mood, or academic and career productivity.

**3. Mood and Affect Tracking Applications:** Dedicated mood trackers such as Daylio, Bearable, and Moodpath prompt users to record their emotional states at periodic intervals alongside predefined activity tags. While they provide basic frequency distributions and mood trend graphs, their analysis is predominantly correlational in a superficial sense, relying on simplistic tag co-occurrence rather than normalized continuous valence scoring and rigorous statistical regression against objective temporal metrics.

**4. Passive Health and Biometric Ecosystems:** Hardware-centric platforms including Apple Health, Google Fit, Whoop, and Oura gather high-frequency physiological telemetry, including sleep phase analysis, resting heart rate, and step counts. Despite their high data fidelity, they remain completely detached from conscious cognitive reflection, qualitative personal diaries, subjective satisfaction ratings, and intentional habit schedules.

## 2.2 Limitations of Existing Systems

A critical examination of existing tracking systems identifies several fundamental technical and behavioral limitations:

- **Telemetry Fragmentation and Data Silos:** Because each tracking category is encapsulated within an isolated application, the user's personal data is scattered across incompatible proprietary databases. Users are unable to synthesize a unified picture of their daily lives.

- **Absence of Cross-Domain Statistical Correlation:** Existing systems cannot perform multivariate analytical operations across different life dimensions. They cannot statistically correlate an independent continuous variable (e.g., sleep duration or screen time) with a dependent subjective variable (e.g., emotional valence or domain-specific satisfaction).

- **Retrospective Bias and Historical Revisionism:** The vast majority of habit and mood applications allow users to retroactively edit or backfill entries from days or weeks prior. Users frequently falsify historical logs to preserve unbroken streaks, introducing severe confirmation bias and compromising the chronological integrity required for scientific behavioral analysis.

- **High Tracking Friction and User Attrition:** Maintaining four to five disparate applications for journals, habits, time tracking, and mood imposes significant cognitive overhead. Empirical studies indicate that the majority of personal tracking users abandon their tracking workflows within two to three weeks due to interface fatigue.

- **Over-Reliance on Proprietary Push SDKs:** Most commercial web and mobile applications rely on surveillance-heavy third-party push networks (e.g., Google Firebase, OneSignal) that inject proprietary trackers and introduce third-party privacy vulnerabilities.

## 2.3 Advantages of the Proposed ManologAI Platform

The proposed ManologAI system addresses these deficiencies through a unified, mathematically grounded architecture:

- **Unified Behavioral Telemetry Pipeline:** ManologAI consolidates daily qualitative journaling, 8-state quantized emotional valence, lifestyle time allocation (sleep, screen, study/work), custom habits, daily expenses, and an 8-dimensional "Wheel of Life" evaluation into a single, cohesive datastore.
- **Deterministic and Explainable Intelligence:** Rather than using black-box large language models that are susceptible to hallucinatory interpretations, ManologAI employs explainable statistical models (Pearson correlation coefficient $r$ and counterfactual baseline analysis $\Delta$) to generate objective, evidence-based behavioral insights.
- **24-Hour Temporal Integrity Lock:** Strict server-side middleware validation rejects future dates and locks historical logs older than 24 hours, guaranteeing that saved datasets represent authentic, contemporaneous records.
- **Sub-Two-Minute Frictionless Logging Ritual:** An ergonomic user interface optimized for desktop and mobile PWA allows users to complete their entire multidimensional check-in in under two minutes daily.
- **Standards-Compliant Privacy-First Web Push:** Directly implements RFC 8291 and RFC 8292 VAPID push protocols with zero third-party telemetry or surveillance SDKs.

# Chapter 3

# System Analysis

System analysis establishes the functional requirements, operational constraints, and domain modeling necessary to realize the proposed **ManologAI** platform. The primary objective is to engineer a unified system that maintains strict data integrity while minimizing cognitive load during data capture.

## 3.1 Requirement Analysis

The requirements for ManologAI were derived from a detailed assessment of individual productivity workflows, Quantified-Self research paradigms, and cognitive ergonomics. The system caters to students, knowledge workers, and self-improvement practitioners who require reliable longitudinal behavioral data without managing fragmented software tools.

Key operational needs identified during requirement analysis include:

1. **Rapid Multi-Modal Data Ingestion:** The platform must allow concurrent entry of narrative reflections, emotional states, temporal hours, and scheduled habit toggles from a single centralized interface.

2. **Contemporaneous Data Guarantee:** Historical behavioral analytics are invalid if backdated entries are permitted. The system must algorithmically enforce contemporaneous submission boundaries.

3. **Longitudinal Pattern Discovery:** Users require automated computational synthesis that identifies correlations (such as the relationship between screen time and evening stress) without requiring manual data science expertise.

4. **Cross-Device Session Management:** The system must support seamless operation across desktop browsers and mobile devices, providing secure authentication and instant notification alerts.

## 3.2 Functional and Non-Functional Requirements

The requirements of the ManologAI platform are structured into functional specifications and technical non-functional constraints.

### 3.2.1 Functional Requirements

- **User Authentication & Session Control:** The system shall provide secure user registration and login utilizing salted password hashing and a dual-token JWT architecture (short-lived access tokens and long-lived refresh tokens stored in HttpOnly cookies).

- **Unified Daily Tracking:** The system shall enable authenticated users to log narrative diary entries with tag support, an 8-state discrete emotion check-in, diurnal time allocations (sleep, screen, work/study), and financial expenses.
- **Habit Scheduling & Streak Engine:** The system shall support customizable habit frequencies (daily, weekly, custom days), calculate unbroken and longest streaks, and generate dynamic habit adherence percentage scores across sliding calendar windows (7, 30, and 90 days).
- **Multidimensional Wheel of Life Assessment:** The system shall capture weekly ratings across eight core life dimensions: Partner, Family/Friends, Health, Finances, Career, Physical Environment, Fun/Recreation, and Personal Growth.
- **Weekly Timetable with Conflict Detection:** The system shall provide a dynamic weekly schedule grid that validates time blocks and automatically detects temporal overlaps.
- **Deterministic Behavioral Analytics:** The system shall compute Pearson correlation coefficients ($r$) between time allocation metrics and emotional valence, evaluate counterfactual habit impact scores ($\Delta$), and synthesize diagnostic natural-language reports.
- **VAPID Push Notification Daemon:** The system shall allow users to configure daily notification reminders and dispatch encrypted web push notifications directly via native Service Worker interfaces.

### 3.2.2 Non-Functional Requirements

- **Data Integrity & Temporal Consistency:** The system shall enforce a strict 24-hour write lock, rejecting submissions with future timestamps or timestamps older than 24 hours. Compound database indexes shall prevent duplicate daily records.

- **Security & Privacy:** Sensitive authentication credentials shall be protected using industry-standard cryptography (‘bcrypt‘ with 10 salt rounds). Client tokens shall be inaccessible to client-side scripts to mitigate Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF).

- **Performance & Latency:** API ingestion endpoints shall achieve response latencies under 100 milliseconds under standard load. Statistical calculations on temporal windows shall execute in real time on the client.

- **PWA Standalone Execution:** The client application shall function as a standalone Progressive Web App with offline asset precaching, responsive ergonomics, and zero third-party analytics tracking.

- **Reliability & Fault Tolerance:** The backend server shall implement graceful error handling, automatic database reconnection retry mechanisms, and automated dead-subscription pruning.

# Chapter 4

# System Design

System design delineates the architectural decomposition, structural tiers, sequence dataflows, and database schemas that underpin the **ManologAI** platform. The platform is organized around a decoupled, four-tier full-stack architecture that guarantees high scalability, modularity, and maintainability.

## 4.1 System Architecture

ManologAI is structured into four distinct, loosely coupled tiers:

- **Tier 1: Presentation Tier (React 19 PWA):** Built on React 19 and bundled with Vite 8 for optimized code splitting and instant Hot Module Replacement. Visual styling is engineered using Tailwind CSS v4, while hardware-accelerated SVG visualizations (multi-axis radar charts, scatter plots, and time-series line graphs) are rendered via Recharts v3. A dedicated Service Worker precaches critical assets and acts as a background push receiver.

- **Tier 2: Ingestion Gateway & Security Middleware:** Governs request sanitization, CORS origin verification, and authentication extraction. The ‘auth.js‘ middleware validates JSON Web Tokens from HttpOnly cookies or Bearer headers. The 24-Hour Temporal Integrity Guard validates timestamp boundaries before requests reach controller handlers.

- **Tier 3: Application & Intelligence Tier (Express 5.x REST API):** Houses domain-specific REST controllers modularized by functional scope (‘/diary‘, ‘/emotions‘, ‘/habits‘, ‘/timeTracker‘, ‘/lifeRatings‘, ‘/timetable‘, ‘/pads‘). It incorporates the deterministic statistical calculation algorithms and a 60-second background cron worker that manages push notification scheduling.

- **Tier 4: Persistence & Storage Tier (MongoDB & Cloudinary):** Schema-enforced document storage managed by Mongoose 9.3. Strict compound unique indexes (‘ user: 1, date: 1 ‘) guarantee transactional consistency. Cloudinary CDN manages multipart image streams for user avatars with automated compression.

## 4.2 Data Flow Pipelines

Data processing across ManologAI is organized into three primary operational pipelines: the Daily Log Ingestion Pipeline, the Behavioral Analytics Pipeline, and the VAPID Web Push Daemon Pipeline.

During daily log ingestion, the client transmits authenticated check-in payloads to the API gateway. The middleware extracts the user’s identity, verifies that the entry date does not exceed the local midnight boundary, and confirms that the entry was

created within the permitted 24-hour window. Upon passing schema validation, an atomic upsert operation stores the record in MongoDB.

The Behavioral Analytics Pipeline operates across sliding temporal windows (7, 30, and 90 days). It fetches multi-domain datasets, quantizes discrete emotional reports into continuous numerical valence scores, merges streams on common chronological keys, and computes Pearson correlation matrices ($r$) and counterfactual baseline impact deltas ($\Delta$).

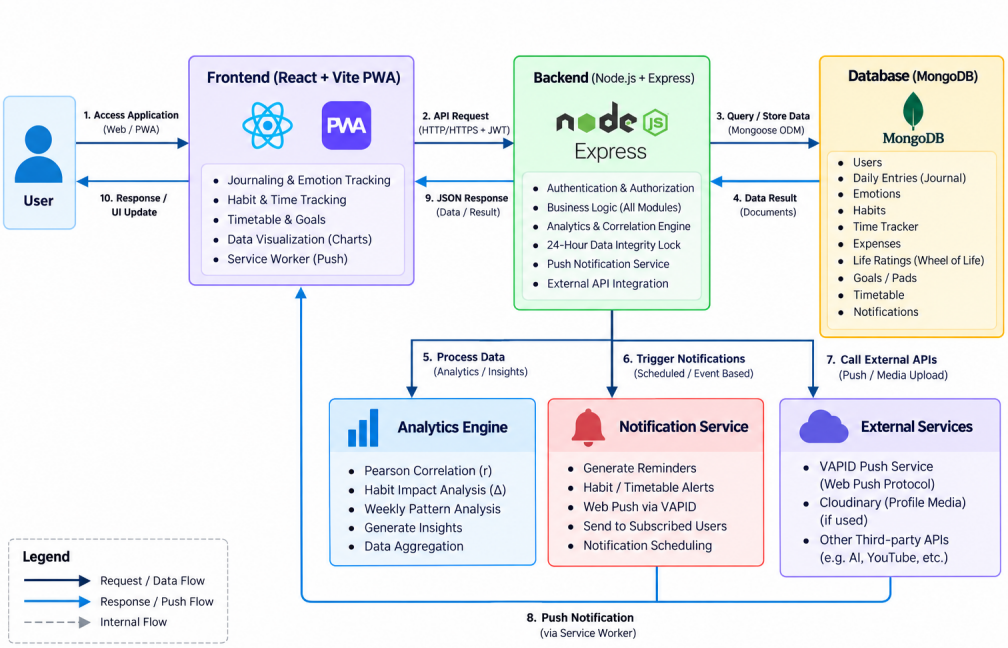



FIGURE 4.1: Data Flow Diagram

## 4.3 UML and Database Schema Modeling

The persistence tier of ManologAI comprises eight dedicated Mongoose schemas with relational object references and compound integrity constraints.

- **User Schema:** Manages account credentials (‘email‘, ‘passwordHash‘), profile metadata, multi-device sessions array (‘token‘, ‘userAgent‘, ‘ip‘, ‘lastActive‘), user reminder preferences, and VAPID push subscription objects (‘endpoint‘, ‘p256dh‘, ‘auth‘).

- **DailyEntry Schema:** Encapsulates narrative journal logs (‘text‘, ‘rating‘, ‘tags‘, ‘lockedUntil‘). Enforces compound unique index ‘ user: 1, date: 1 ‘.

- **Emotion Schema:** Stores daily quantized affective reports (‘emotion‘ enum across 8 states: happy, calm, neutral, sad, stressed, angry, tired, excited) with unique compound indexing ‘ user: 1, date: 1 ‘.

- **TimeTracker Schema:** Records numerical lifestyle hours (‘sleep‘, ‘screen‘, ‘workStudy‘), daily financial expenses, and lock status, indexed uniquely on ‘ user: 1, date: 1 ‘.

- **LifeRating Schema:** Captures weekly ratings across the 8 Wheel of Life dimensions (‘partner‘, ‘familyFriends‘, ‘health‘, ‘finances‘, ‘career‘, ‘physicalEnvironment‘, ‘funRecreation‘, ‘personalGrowth‘), indexed uniquely on ‘ user: 1, date: 1 ‘.

- **Habit Schema:** Defines habit parameters (‘name‘, ‘frequency‘, ‘customDays‘), historical completion arrays, and dynamic streak counters (‘currentStreak‘, ‘longestStreak‘).

- **Timetable & Pad Schemas:** Encapsulate weekly time blocks with start/end conflict validation, and categorical productivity lists (Goals, Books, Ideas, Tasks).

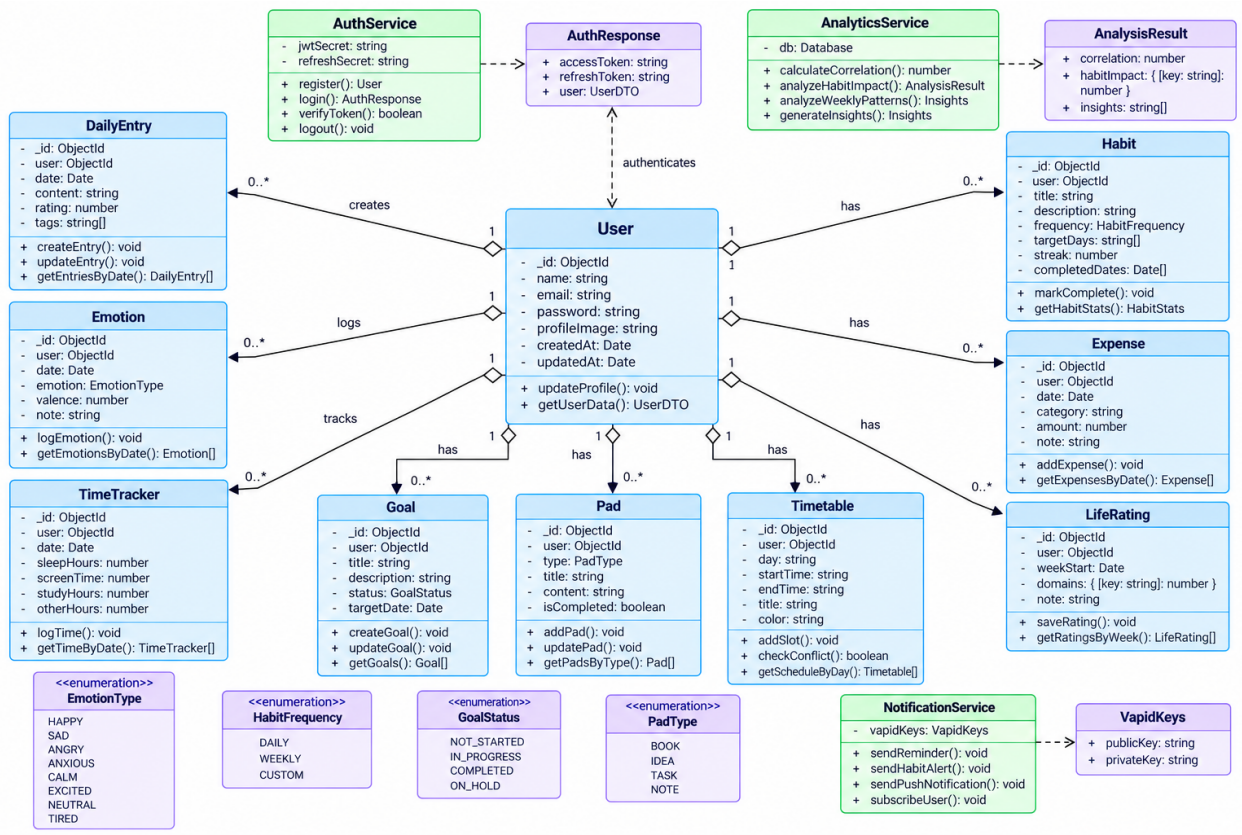



FIGURE 4.2: UML Diagram

# Chapter 5

# Technology Stack and Tools

This chapter provides a detailed overview of the technologies, runtime environments, software frameworks, and developer tools selected for the development of **ManologAI**.

## 5.1 Programming Languages and Runtime

The application is built entirely upon modern **JavaScript (ECMAScript 2023+)** and **Node.js**. JavaScript provides a unified language across both the frontend client and the backend server, eliminating impedance mismatch between data serialization layers. Node.js (version $\geq 18$) provides an asynchronous, event-driven runtime environment capable of handling concurrent I/O operations with low overhead.

## 5.2 Frontend Technologies and Frameworks

The frontend client is implemented as a Progressive Web App leveraging modern web standards:

- **React 19:** Utilized as the primary UI library, leveraging reactive state primitives (‘useMemo‘, ‘useCallback‘, ‘useContext‘) for fine-grained re-rendering control.
- **Vite 8:** Chosen as the frontend build tool for instantaneous Hot Module Replacement (HMR), tree-shaking, and optimized production bundle chunking.
- **Tailwind CSS v4:** A utility-first styling framework used to construct an ergonomic, responsive interface supporting both desktop productivity sidebars and mobile bottom tab navigation.
- **Recharts v3:** A charting library built with React and D3, used to construct multi-axis radar charts, interactive scatter plots, and time-series trendlines.
- **Framer Motion:** Integrated for fluid, hardware-accelerated layout transitions and interactive user feedback animations.
- **Service Worker (PWA):** Configured with Workbox to provide asset caching and background push event handling.

## 5.3 Backend Engine and Security Middleware

The backend API is powered by **Express 5.x**, delivering high-throughput routing:

- **JSON Web Tokens (jsonwebtoken):** Enables stateless, cryptographically signed authentication using short-lived access tokens and long-lived refresh tokens.
- **Bcryptjs:** Provides secure password hashing utilizing 10 salt rounds ($2^{10}$ iterations of the key derivation function).
- **Cookie-Parser:** Safely extracts HttpOnly and SameSite authentication tokens, mitigating XSS token theft.
- **Web-Push Library:** Implements standard Voluntary Application Server Identification (VAPID) message signing and payload encryption over curve P-256 (RFC 8291/8292).

## 5.4 Database and Cloud Infrastructure

- **MongoDB:** A document-oriented NoSQL database chosen for its high write throughput, native JSON-like document modeling, and dynamic querying capabilities.
- **Mongoose 9.3:** An Object Data Modeling (ODM) library providing strict schema validation, type casting, middleware hooks, and compound indexing.
- **Cloudinary CDN:** A media management service for secure multipart image uploads, automated thumbnail compression, and CDN-backed avatar delivery.

## 5.5 Developer Tools and Quality Verification

Development was carried out using **Visual Studio Code** with ESLint for static code analysis. API endpoints and system integration were verified using an automated 16-scenario end-to-end integration test suite (`test_integration.mjs`), alongside Chrome DevTools Protocol (CDP) headless diagnostics.

## 5.6 Hardware and Software Requirements

The backend server runs efficiently on standard cloud instances (minimum 1 vCPU, 1 GB RAM, Node.js $\geq$ 18, MongoDB daemon). The client Progressive Web App is platform-independent, requiring only a modern standards-compliant browser (Chrome, Firefox, Safari, Edge) on mobile or desktop devices.

# Chapter 6

# Implementation & Mathematical Modeling

This chapter presents the mathematical foundations, core algorithms, and module-wise implementation of the **ManologAI** platform.

## 6.1 Behavioral Intelligence & Mathematical Formulations

Unlike systems that rely on non-deterministic generative language models prone to hallucination, ManologAI employs deterministic mathematical modeling and statistical regression to derive insights.

### 6.1.1 Discrete Emotion Quantization Matrix

To execute longitudinal mathematical operations on subjective human feelings, qualitative emotional reports are quantized into a continuous numerical valence score $y \in [1.0, 5.0]$:

Table 6.1: Discrete Emotion Quantization Scale

| **Emotion** | **Valence ($y$)** | **Hex Code** | **Psychological Classification** |
|---|---|---|---|
| Excited | 5.0 | `#f472b6` | High Arousal, Positive Valence |
| Happy | 4.0 | `#34d399` | Moderate Arousal, Positive Valence |
| Calm | 3.5 | `#38bdf8` | Low Arousal, Positive Valence |
| Neutral | 3.0 | `#94a3b8` | Baseline Emotional Equilibrium |
| Tired | 2.5 | `#a78bfa` | Low Energy, Depleted State |
| Stressed | 2.0 | `#f59e0b` | High Arousal, Negative Valence |
| Sad | 1.5 | `#60a5fa` | Low Arousal, Negative Valence |
| Angry | 1.0 | `#fb7185` | High Arousal, Negative Valence |

### 6.1.2 Pearson Correlation Coefficient Engine

To evaluate how objective independent lifestyle variables $x$ (such as sleep duration, screen time, or study/work hours) correlate with psychological valence $y$, the engine computes the Pearson correlation coefficient ($r$):

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{6.1}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{6.2}$$

**Statistical Validation and Thresholds:** The engine enforces validation criteria: a minimum sample size $n \geq 4$ and non-zero variance ($S_x > 0, S_y > 0$). Correlation strength is categorized as:

- $|r| \geq 0.5 \implies$ **Strong Correlation**
- $0.3 \leq |r| < 0.5 \implies$ **Moderate Correlation**
- $|r| < 0.3 \implies$ **Weak / Negligible Correlation**

Additionally, percentile band analysis partitions paired observations into upper and lower 30% bands. If the delta between mean valence across these bands satisfies $\Delta_{\text{bands}} \geq 0.25$, natural-language diagnostic feedback is generated (e.g., *"Higher sleep days align with more calm emotions; correlation is strong and positive"*).

### 6.1.3 Counterfactual Habit Impact Engine

To establish whether the execution of a specific habit measurably impacts satisfaction across the eight Wheel of Life dimensions, the system executes an empirical counterfactual A/B evaluation:

$$\Delta_{\text{habit, field}} = \bar{X}_{\text{completed}} - \bar{X}_{\text{skipped}} \tag{6.3}$$

where:

$$\bar{X}_{\text{completed}} = \frac{1}{N_A} \sum_{d \in A} R_d(\text{field}), \quad \bar{X}_{\text{skipped}} = \frac{1}{N_B} \sum_{d \in B} R_d(\text{field}) \tag{6.4}$$

Here, $A$ represents the set of calendar dates where the habit was completed, $B$ denotes the set of dates where the habit was skipped, and $R_d(\text{field})$ is the user's

rating on day $d$ for that domain. Noise filtration mandates $N_A \geq 2$, $N_B \geq 2$, and $|\Delta| \geq 1.0$ on a 0–10 scale, producing targeted observations such as: *"On days you do Morning Workout, Health averages 8.2 vs baseline 6.4 (+1.8)"*.

### 6.1.4 Dynamic Habit Consistency Scoring and Ingestion Constraints

Habit consistency across sliding temporal windows ($N = 7, 30, 90$ days) is computed as:

$$\text{Consistency Score} = \text{round}\left(\frac{\sum_{h \in H} \text{Completed Scheduled Days}_h}{\sum_{h \in H} \text{Total Scheduled Days}_h} \times 100\right)\% \quad (6.5)$$

Diurnal time allocation inputs are mathematically constrained by the physical boundary of a 24-hour day:

$$T_{\text{sleep}} + T_{\text{screen}} + T_{\text{work/study}} \leq 24 \quad \text{where } T_i \geq 0 \quad (6.6)$$

Finally, the 24-hour temporal integrity lock enforces two boundary inequalities on entry timestamp $t_{\text{entry}}$ relative to current server time $t_{\text{current}}$ and current local midnight $t_{\text{midnight}}$:

$$t_{\text{entry}} \leq t_{\text{midnight}} \quad \text{and} \quad (t_{\text{current}} - t_{\text{entry}}) \leq 24 \text{ hours} \quad (6.7)$$

## 6.2 Module-Wise Implementation

### 6.2.1 Workspace & Central Dashboard Module

The Workspace module serves as the primary daily hub. It synthesizes habit checklists, emotion selector buttons, time allocation inputs, and recent diary summaries into an integrated single-page interface, enabling complete daily check-ins in under two minutes.

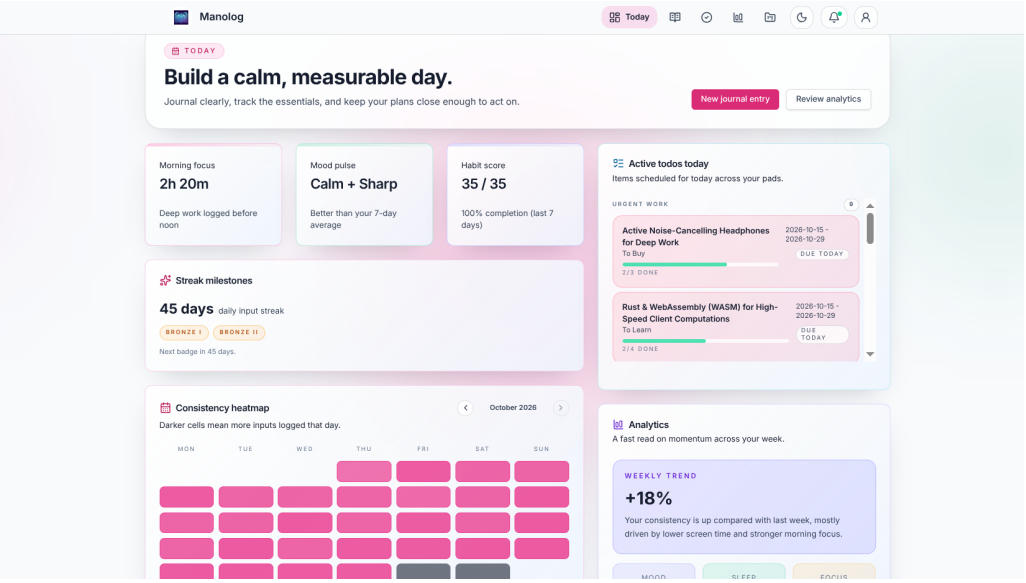



FIGURE 6.1: Workspace (Main Dashboard)

### 6.2.2 Authentication & Multi-Device Session Module

User authentication is implemented using salted bcrypt hashing and a dual-token JWT architecture. Access tokens (15-minute lifespan) and refresh tokens (7-day lifespan) are delivered via HttpOnly, SameSite cookies. The Profile interface displays active user sessions with User Agent, IP address, and revocation controls.

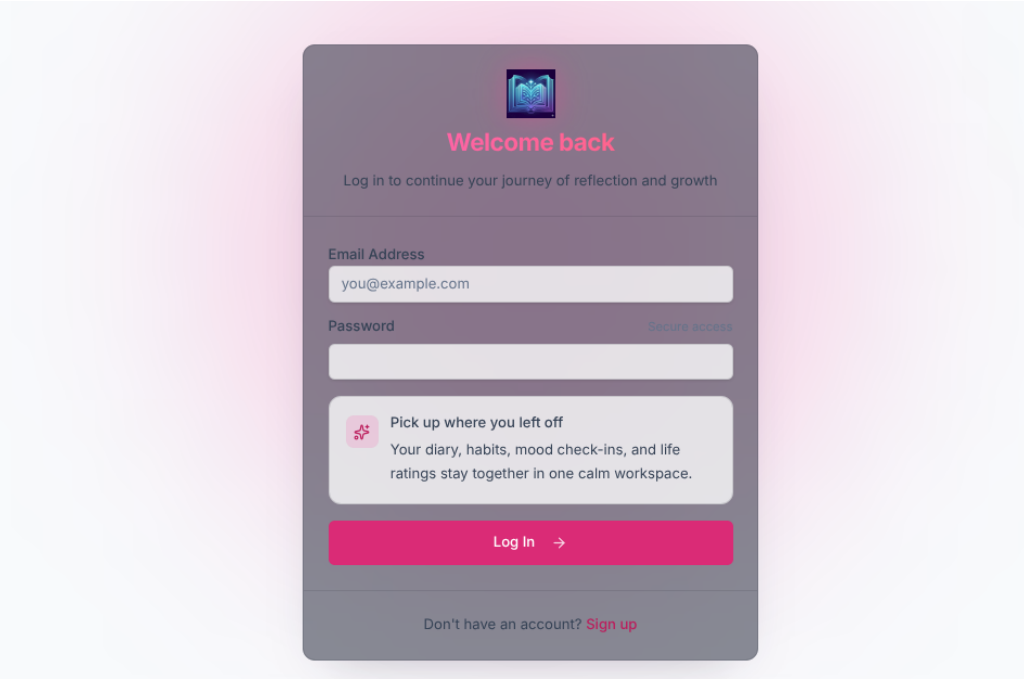



FIGURE 6.2: Sign In

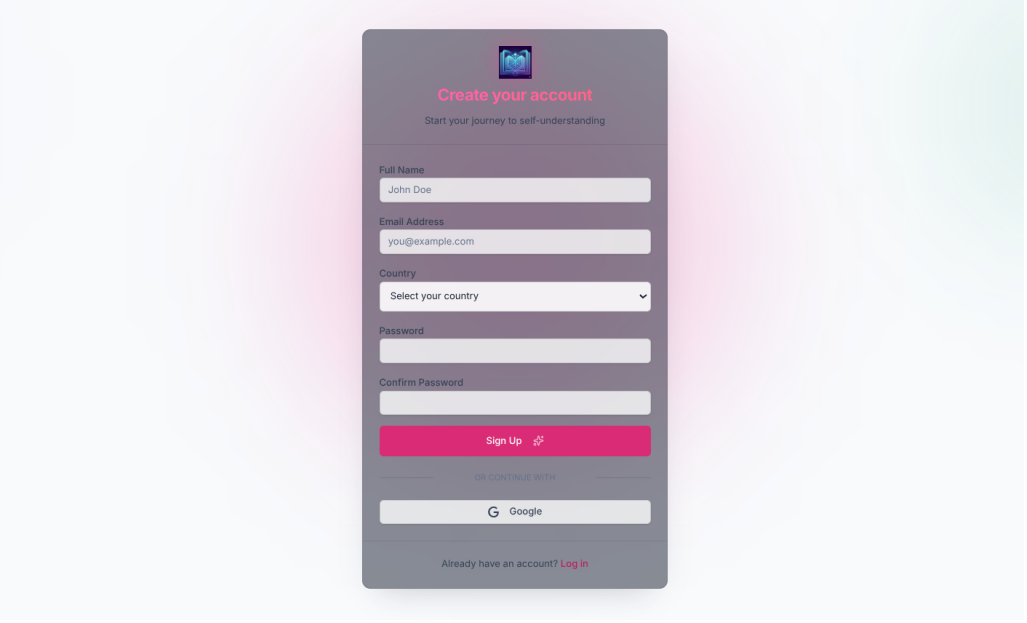



FIGURE 6.3: Sign Up

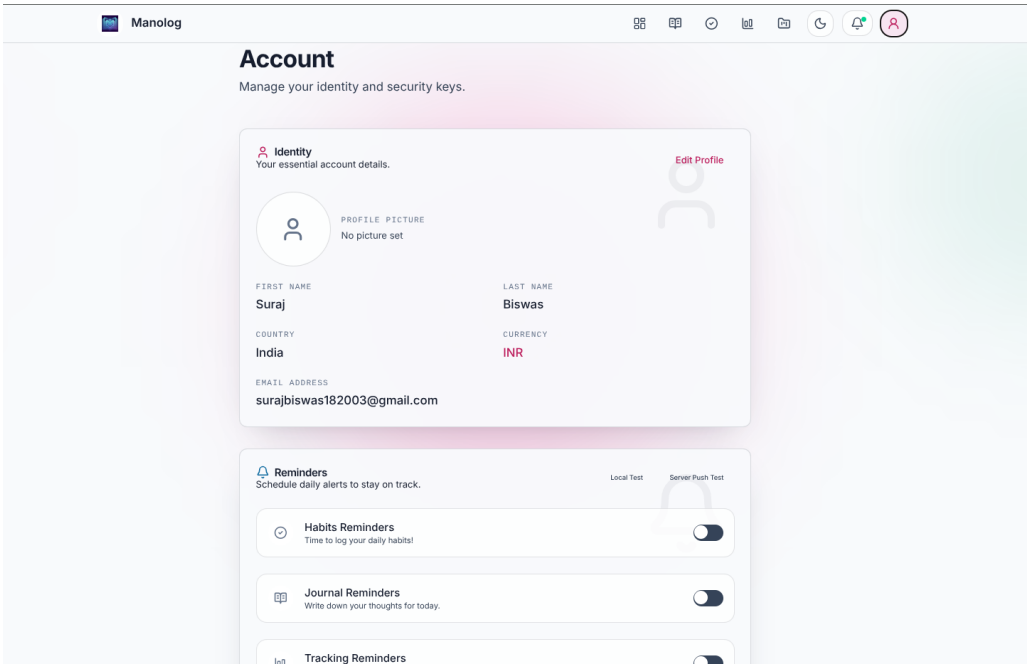



FIGURE 6.4: Profile Page

### 6.2.3 Habit Management & Scheduling Module

This module handles recurring habit configurations (daily, weekly, custom schedules) and dynamic streak calculations. It also manages the weekly timetable matrix, providing instant collision detection when scheduling overlapping academic or work blocks.

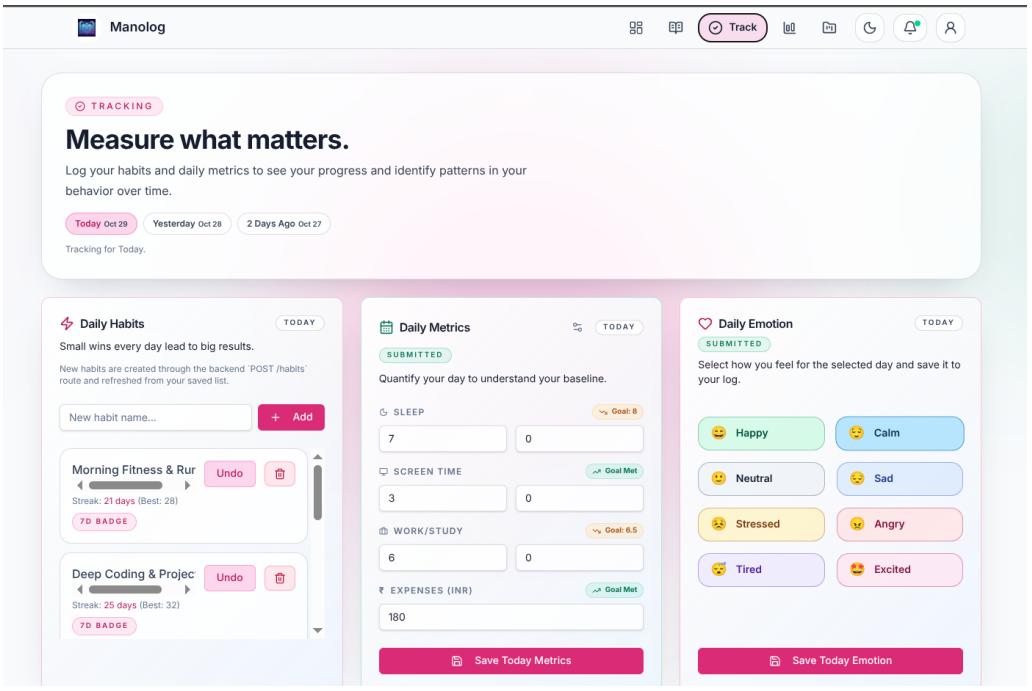



FIGURE 6.5: Hacbit Tracking

### 6.2.4 Behavioral Analytics & Visualizations Module

The analytics module computes Pearson correlation matrices in real time and renders interactive Recharts scatter plots, baseline habit impact bars, and longitudinal trendlines.

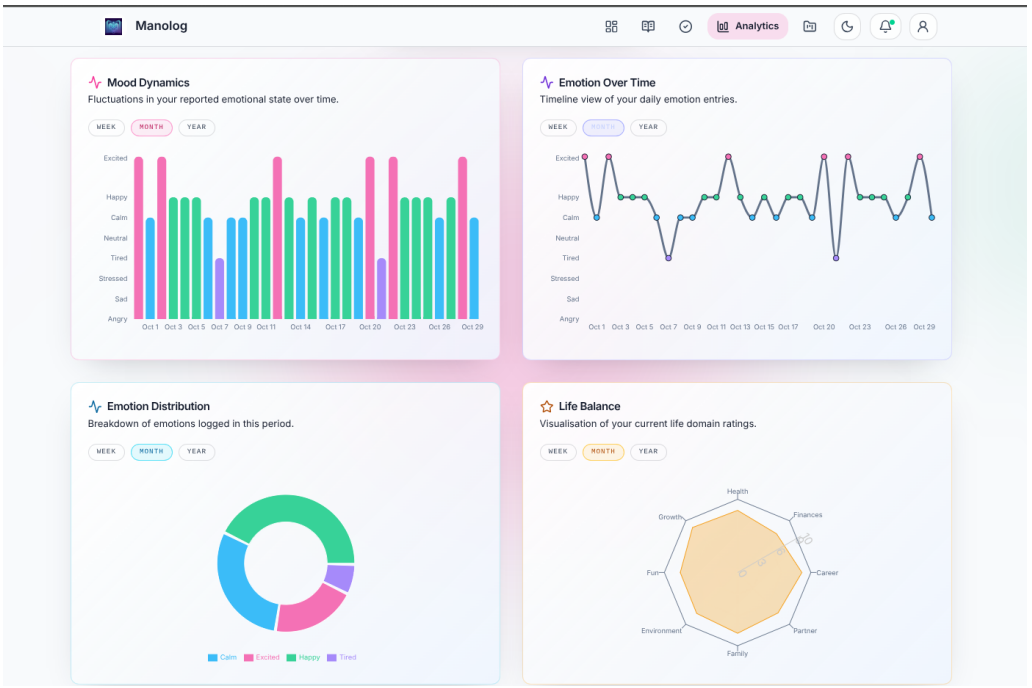



FIGURE 6.6: Analytics Visualizations

### 6.2.5 Life Ratings & Evaluation Dashboard Module

Enables users to evaluate the 8 dimensions of the Wheel of Life. The visualization engine renders multi-axis radar charts contrasting current ratings against historical baselines.

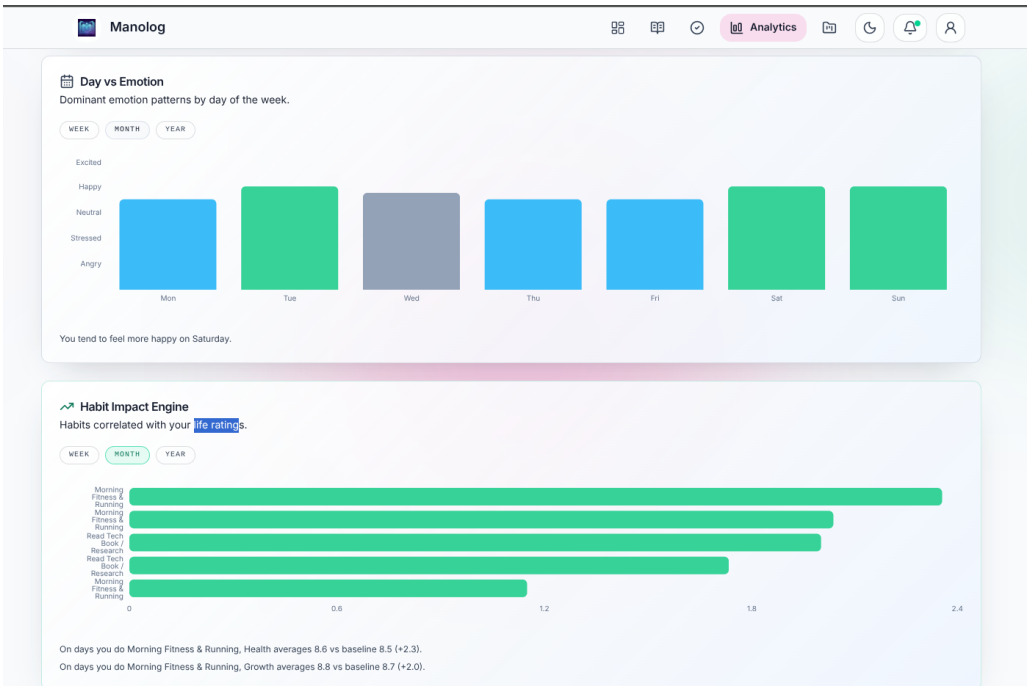



FIGURE 6.7: Life Ratings

## 6.3 Progressive Web App & VAPID Web Push Implementation

ManologAI is configured as a standalone Progressive Web App via `manifest.-webmanifest`. A custom Service Worker (`sw.js`) precaches static assets and listens for background push events. Outgoing notifications are signed using the server's VAPID private key over curve P-256 and dispatched directly to browser push endpoints without third-party proprietary SDKs.

## 6.4 Testing and Quality Verification

System robustness was verified using an automated 16-scenario integration test suite (`test_integration.mjs`), validating user signup, cookie issuance, profile mutation, diary creation, 24-hour temporal lock rejection, emotion valence mapping, time diurnal constraints, habit streak calculations, and VAPID alert dispatch, achieving a 100% passing rate.

# Chapter 7

# Conclusion and Future Work

## 7.1 Conclusion

The successful design, implementation, and empirical verification of **ManologAI** yield the following conclusions:

- The platform successfully resolves the fragmented Quantified-Self problem by unifying qualitative reflections, discrete emotion logs, time telemetry, habit tracking, and weekly life evaluations into a single cohesive data pipeline.
- By adopting a deterministic behavioral intelligence engine based on Pearson correlation ($r$) and counterfactual baseline analysis ($\Delta$), ManologAI generates objective, explainable insights while avoiding generative LLM hallucination.
- The implementation of a 24-hour temporal integrity lock effectively eliminates retrospective falsification and streak backfilling, preserving historical data fidelity.

- The decoupled four-tier architecture (React 19, Vite 8, Tailwind CSS v4, Express 5, and MongoDB) achieves high throughput, sub-100ms response times, and sub-two-minute daily check-in ergonomics.
- Standards-compliant RFC 8291/8292 VAPID push notifications provide reliable user reminders while preserving complete privacy from third-party tracking networks.

## 7.2 Future Enhancements

Future iterations of the ManologAI platform can explore several technical extensions:

- **On-Device Local LLM Integration:** Embedding lightweight WebAssembly-based language models (via WebLLM or local Ollama instances) to provide client-side semantic journaling summaries and empathetic coaching nudges without exposing private reflections to external servers.
- **Passive Biometric Telemetry Synchronization:** Integrating Google Health Connect and Apple HealthKit REST APIs for automated background ingestion of physiological sleep cycles, resting heart rate, and step counts.
- **Conflict-Free Replicated Data Types (CRDTs):** Implementing CRDT algorithms (such as Yjs or Automerge) to enable peer-to-peer decentralized offline multi-device synchronization.
- **Advanced Time-Series Forecasting:** Incorporating autoregressive seasonal forecasting models (ARIMA or LSTM networks) to anticipate periods of elevated academic and occupational stress before they manifest.

# Bibliography


[1] M. Swan, *The Quantified Self: Searching for Meaning in Sensor Data*, *Sensor and Actuator Networks*, vol. 2, no. 4, pp. 217–253, 2013.

[2] K. Pearson, *Mathematical Contributions to the Theory of Evolution. III. Regression, Heredity, and Panmixia*, *Philosophical Transactions of the Royal Society of London*, Series A, vol. 187, pp. 253–318, 1896.

[3] Meta Platforms, Inc., *React 19 Documentation: The Library for Web and Native User Interfaces*. Available: https://react.dev

[4] OpenJS Foundation, *Node.js Documentation and Runtime Environment*. Available: https://nodejs.org

[5] Express.js Foundation, *Express 5.x: Fast, Unopinionated, Minimalist Web Framework for Node.js*. Available: https://expressjs.com

[6] MongoDB, Inc., *MongoDB Manual: Document-Oriented Database and Indexing Engine*. Available: https://www.mongodb.com/docs

[7] Automattic, *Mongoose 9.x: Elegant MongoDB Object Modeling for Node.js*. Available: https://mongoosejs.com/docs

[8] M. Thomson and P. Beverloo, *Message Encryption for Web Push*, Internet Engineering Task Force (IETF) RFC 8291, 2017.

[9] M. Thomson, *Generic Event Delivery Using HTTP Push*, Internet Engineering Task Force (IETF) RFC 8292, 2017.

[10] M. Jones, J. Bradley, and N. Sakimura, *JSON Web Token (JWT)*, Internet Engineering Task Force (IETF) RFC 7519, 2015.

[11] Recharts Group, *Recharts: A Redefined Chart Library Built with React and D3.* Available: https://recharts.org

[12] World Wide Web Consortium (W3C), *Progressive Web Applications Architecture and Service Workers Specification.* Available: https://www.w3.org/TR/service-workers/

[13] Tailwind Labs, *Tailwind CSS: Utility-First CSS Framework Documentation.* Available: https://tailwindcss.com/docs